DATA STRUCTURE

LAB PROGRAMS

21. Quick Sort

```
#include <stdio.h>
int partition(int arr[], int low, int high) {
    int pivot = arr[high];
    int i = low - 1;
    for(int j=low;j<high;j++) {
        if(arr[j] < pivot) {
            i++;
            int temp = arr[i];
            arr[i] = arr[j];
            arr[j] = temp;
        }
    }
    int temp = arr[i+1];
    arr[i+1] = arr[high];
    arr[high] = temp;
    return i+1;
}
void quickSort(int arr[], int low, int high) {
    if(low < high) {
        int pi = partition(arr,low,high);
        quickSort(arr,low,pi-1);
        quickSort(arr,pi+1,high);
    }
}
```

```
int main() {
    int arr[] = {10,7,8,9,1,5};
    int n = 6;
    quickSort(arr,0,n-1);
    printf("Sorted Array:\n");
    for(int i=0;i<n;i++)
        printf("%d ",arr[i]);
    return 0;
}
```

Output

```
Sorted Array:
1 5 7 8 9 10

=== Code Execution Successful ===
```